Project Title: Dynamic Data Cleaning Automation Tool using Python

□ Project Overview

Developed a desktop-based data cleaning application using Python and Tkinter that automates common data preprocessing tasks on Excel files. The tool provides an interactive graphical interface where users can select datasets, apply cleaning operations, and export cleaned data effortlessly.

🎯 Objective

The main goal of this project is to simplify and automate repetitive data cleaning tasks such as removing duplicates, handling blank rows, and formatting text, making data preprocessing faster and more user-friendly for analysts.

🛠️ Tools & Technologies Used

- Python
- Tkinter (GUI Development)
- Pandas (Data Manipulation)
- Openpyxl (Excel File Handling)
- OS Module (File Management)

⚙️ Key Features

- 🗁 File selection using a graphical file browser
- 🗁 Output folder selection for saving processed data
- □ Data cleaning options:
    - Remove duplicate rows
    - Remove blank rows
    - Trim leading/trailing spaces
    - Convert text columns to Title Case
- 🖫 Automatically saves cleaned Excel file with updated name
- ⚠ Error handling with user-friendly message boxes

- 📄 Simple and interactive GUI interface

🔄 Workflow

1. User selects an input Excel file
2. Chooses output folder
3. Selects required cleaning operations via checkboxes
4. Clicks “Clean Data” button
5. Tool processes the dataset using Pandas
6. Cleaned file is saved automatically in the selected location

---

☑️ Outcome

- Reduced manual effort in data cleaning tasks
- Improved speed and accuracy of preprocessing
- Built a user-friendly automation tool for non-technical users
- Strengthened practical understanding of Python, Pandas, and GUI development

---

💡 Learning Highlights

- Hands-on experience in building GUI applications using Tkinter
- Practical implementation of Pandas for real-world data cleaning
- Understanding of file handling and automation in Python
- Improved problem-solving and software development skills

🚀 Future Enhancements

- Add data preview feature before cleaning
- Include data visualization dashboard
- Convert into a web-based application using Streamlit
- Package as a standalone desktop application (.exe)